4. **Robustness and fairness perturbations.** Across all scenarios, we observe strong correlations between accuracy, robustness, and fairness, where robustness and fairness metrics consider worst-case accuracy over a set of perturbations (e.g. typos for robustness, dialect alteration for fairness)—see §4.5: METRICS-ROBUSTNESS, §4.6: METRICS-FAIRNESS for more details. While there is a strong correlation between accuracy and fairness (Figure 24, Figure 25), we do observe trade-offs where the most accurate model is not the most robust or most fair. We also see serious drops in some cases: for example, on **NarrativeQA**, TNLG v2 (530B) precipitously drops from 72.6% standard accuracy (i.e. the third-most accurate model) to 38.9% accuracy in the presence of robustness perturbations.[8]

5. **Performance disparities.** When we have access to demographic metadata, we generally see consistent performance disparities for all models. As an example of racialized dialect disparities, OPT (175B) is the most accurate model on **TwitterAAE** but its accuracy degrades from 1.506 bits per byte for White English to 2.114 bits per byte for African American English (lower is better).[9]

6. **Generative harms.** We find that the biases and toxicity in model generations are largely constant across models and low overall on average for the core scenarios (Figure 24). However, note that even low levels of bias or toxicity could cause non-trivial social harm, and targeted evaluations are needed to obtain a more detailed characterization (§5.6: TARGETED-BIAS, §5.7: TARGETED-TOXICITY).

7. **Accuracy vs. efficiency.** We do not see a strong trade-off between accuracy and efficiency (which depends on both the model architecture and the hardware, see §4.9: METRICS-EFFICIENCY) across all 30 models (Figure 24). For each family of models (e.g. different size variants of GPT-3), we find that as models become larger, accuracy consistently improves but with higher training and inference cost.[10] Overall, we observe that only a subset of all models (across model families) are on the accuracy-efficiency Pareto frontier for each scenario.

8. **Question answering.** Across the 9 core question answering scenarios (§3.3: QUESTIONANSWERING), we observe significant heterogeneity in results, though text-davinci-002 is the most accurate model for all 9 scenarios.[11] In fact, for 6 of the 9 scenarios, there is no open model among the three most accurate models, as generally they are text-davinci-002, Anthropic-LM v4-s3 (52B), and TNLG v2 (530B) in descending order of accuracy.

9. **Information retrieval.** We consider the classic task of ranking candidate passages given a query (§3.4: INFORMATIONRETRIEVAL). The best-performing models we evaluate outperform classical retrieval methods and under some settings perform comparably to various fine-tuned neural retrievers, while nonetheless trailing the state of the art.[12] Because the number of candidates could be large, we create a LM request per passage, which requires the model to produce calibrated probabilities. Our use of LMs for passage ranking is unorthodox, and computationally intensive in its naive implementation, but we include it as a proof of concept.

10. **Summarization.** **CNN/DailyMail** and **XSUM** have been standard benchmarks for summarization for many years, but we find that automated evaluations on these datasets largely fail to discriminate differences we observed in model quality.

11. **Sentiment analysis.** For sentiment analysis on **IMDB**, many models are quite accurate and well-calibrated with marginal drops on robustness and fairness perturbations, but the contrast sets of Gardner et al. (2020) highlight clear limitations in model robustness (e.g. one of the most accurate models in GLM (130B) drops by more than 8%).[13]

12. **Toxicity detection.** For toxicity detection on **CivilComments**, we find that most models are not particularly accurate: OPT (175B) is one of the most accurate models across all scenarios

[8] See `https://crfm.stanford.edu/helm/v0.1.0/?group=narrative_qa`.
[9] See `https://crfm.stanford.edu/helm/v0.1.0/?group=twitter_aae`.
[10] See `https://crfm.stanford.edu/helm/v0.1.0/?group=core_scenarios#Efficiency`.
[11] See `https://crfm.stanford.edu/helm/v0.1.0/?group=question_answering`.
[12] See `https://crfm.stanford.edu/helm/v0.1.0/?group=information_retrieval`.
[13] See `https://crfm.stanford.edu/helm/v0.1.0/?group=sentiment_analysis` and `https://crfm.stanford.edu/helm/v0.1.0/?group=robustness_contrast_sets`.

(Figure 26), but achieves essentially chance accuracy at 50.1%.[14] Critically, given the importance of fairness in toxicity detection due the disparate impacts of content moderation, we find that most models are similarly accurate for detecting toxicity in comments mentioning Black and White individuals. However, models vary greatly in their robustness: OPT (175B) drops from 51.3% standard accuracy to 8.8% robust accuracy on the Black split, whereas the drop is less precipitous on the White split (50.8% to 24.3%).

13. **Miscellaneous text classification.** For text classification on **RAFT**, we see significant heterogeneity in which models do well on which subsets/tasks.[15] text-davinci-002 is consistently accurate across splits when compared with other models, but performs very poorly on the Systematic Review Inclusion split with an accuracy of 40.8% compared to 97.5% from several models (e.g. GLM (130B)).

14. **Linguistic understanding.** The trends in accuracy for language modeling[16] are quite different from the trends for the core scenarios (Figure 26) . In particular, GPT-NeoX (20B), OPT (175B), BLOOM (176B), GPT-J (6B), and OPT (66B) consistently have the lowest bits-per-byte (lower is better) on **The Pile**, **TwitterAAE**, and **ICE**. In terms of linguistic phenomena, all models perform fairly similarly on **BLiMP** overall, and further perform very similarly even on each of the specific subsets for morphology, syntax, semantics, and syntax-semantics. We see the widest spread on irregular forms (morphology), where surprisingly the models that tend to be the most accurate for core scenarios (i.e. text-davinci-002, TNLG v2 (530B)) are some of the least accurate for irregular forms, perhaps suggesting they have overgeneralized particular linguistic rules.[17]

15. **Knowledge.** text-davinci-002 demonstrates superior performance for all knowledge-intensive evaluations,[18] with a very sizable gap for accuracy on **TruthfulQA** of 62.0% compared to second place of 36.2% from Anthropic-LM v4-s3 (52B).[19] Further, TNLG v2 (530B) shows strong performance on the highly knowledge-intensive **NaturalQuestions** (closed-book) and **WikiFact** scenarios, which generally concurs with the hypothesis that model scale especially contributes to improvements in acquisition of factual knowledge. For example, Anthropic-LM v4-s3 (52B) and TNLG v2 (530B) tend to get very similar accuracies for most scenarios (as suggested by Figure 26), but TNLG v2 (530B) demonstrates a wide margin for these two scenarios (38.5% vs. 28.7% for **NaturalQuestions** (closed-book), 34.3% vs. 22.3% for **WikiFact**).

16. **Reasoning.** For reasoning-intensive scenarios, we find that the code models, especially code-davinci-002, consistently outperform the text models, even on synthetic reasoning scenarios posed in natural language.[20] This gap is made clear in mathematical reasoning: for **GSM8K**, code-davinci-002 achieves an accuracy of 52.1%, where the next best model is text-davinci-002 at 35.0% and no other model surpasses 16%.[21] Further, in addition to code-davinci-002, text-davinci-002 is much more accurate than other text models (e.g. 65.1% accuracy on synthetic reasoning in natural language, whereas the next most accurate text model is OPT (175B) at 29.4% accuracy, and code-davinci-002 has an accuracy of 72.7%).

17. **Memorization of copyrighted/licensed material.** We find that the likelihood of direct regurgitation of long copyrighted sequences is somewhat uncommon, but it does become noticeable when looking at popular books.[22] However, we do find the regurgitation risk clearly correlates with model accuracy: text-davinci-002, davinci (175B), and Anthropic-LM v4-s3 (52B) demonstrate the highest amount of verbatim regurgitation in line with their high accuracies.

[14]See `https://crfm.stanford.edu/helm/v0.1.0/?group=civil_comments`.
[15]See `https://crfm.stanford.edu/helm/v0.1.0/?group=raft`.
[16]See `https://crfm.stanford.edu/helm/v0.1.0/?group=language#Accuracy`.
[17]See `https://crfm.stanford.edu/helm/v0.1.0/?group=blimp#phenomenon:%20irregular_forms`.
[18]See `https://crfm.stanford.edu/helm/v0.1.0/?group=knowledge#Accuracy`.
[19]See `https://crfm.stanford.edu/helm/v0.1.0/?group=truthful_qa`. We note this is especially interesting given the projections of model accuracy by Evans et al. (2022), though we note our results are for 5-shot learning whereas their projections are for 0-shot learning.
[20]See `https://crfm.stanford.edu/helm/v0.1.0/?group=reasoning#Accuracy`.
[21]See `https://crfm.stanford.edu/helm/v0.1.0/?group=gsm`.
[22]See `https://crfm.stanford.edu/helm/v0.1.0/?group=copyright_text`.

18. **Disinformation.** We find that the largest models (particularly text-davinci-002 and Anthropic-LM v4-s3 (52B)) are effective at generating realistic headlines that support a given thesis,[23] but results are more mixed when prompting models to generate text encouraging people to perform certain actions (Table 8).[24]

19. **Targeted biases.** For **BBQ**, text-davinci-002 is the most accurate model by a very wide margin (89.5% accuracy), with the next most accurate models (T0++ (11B), 48.4%; TNLG v2 (530B), 44.9%) being the only other models with accuracies above 40%. We highlight this because we see a very striking relationship on **BBQ** between model accuracy and model bias for ambiguous contexts. These three models, which are the three most accurate, are the only three models with biases in ambiguous contexts that align with broader social biases/discrimination, whereas all other models show biases in the other direction (Figure 40). In other words, we find that for **BBQ** the most accurate models are precisely those that are most concerning for social biases in ambiguous contexts, though the trends in disambiguated contexts are less clear.

20. **Targeted toxicity generation.** For the core scenarios, we observed the rate of toxicity generation was quite low. Honing in on toxicity generation, all models show much stronger tendencies for toxic generations for toxic prompts in **RealToxicityPrompts**, as compared to relatively non-toxic prompts in both **RealToxicityPrompts** and **BOLD**.[25] Understanding how these trends change based on the automated toxicity detection model used (currently PerspectiveAPI), as well as when human judgments from diverse stakeholders are used, is a key area for future work.

21. **Comprehensiveness.** By evaluating under unified conditions extensively, we expose findings lying in plain sight. In other words, while in many cases we are evaluating models that are available publicly on datasets that are available publicly, we nonetheless surface new findings. As an example, we find text-davinci-002 achieves an accuracy of 74.4% `ROUGE-L` on **NarrativeQA**, which sets a new state-of-the-art across all methods to our knowledge, in this case over the strong QA-specialized UnifiedQA-v2 model (67.4% `ROUGE-L`; Khashabi et al., 2022).

22. **Prompting.** All models show significant sensitivity to the formatting of prompt, the particular choice of in-context examples, and the number of in-context examples across all scenarios and for all metrics (see §8.2: prompting-analysis). In this effort, we consistently work towards standardizing these dimensions (e.g. to ensure models are interoperable/performant using the same prompting practices), but current models differ in what prompting decisions would maximize accuracy.[26]

23. **Multiple choice adaptation method.** We find that model performance is extremely sensitive to how multiple choice scenarios are adapted into prompts: for example, accuracy for OPT (175B) on **HellaSwag** is 79.1% when each answer choice is presented in a separate 0-shot prompt (i.e. one of the most accurate models), but drops precipitously to 30.2% (almost random accuracy) when the answer choices are presented jointly in a single 5-shot prompt (i.e. in the format of a multiple-choice exam).[27] Further, even for the same scenario, the adaptation method that maximizes accuracy can differ (and produce qualitatively different results) across models (Figure 33) . This poses a fundamental challenge for what it means to standardize language model evaluation in a fair way across models.

24. **Upstream perplexity and downstream accuracy.** Given the myriad scenarios where LMs could provide value, it would be appealing for many reasons if upstream perplexity on language modeling objectives reliably predicted downstream accuracy. Unfortunately, when making these comparisons across model families, even when using bits-per-byte (BPB; which is more comparable than perplexity), we find this type of prediction does not work well: BPB on **The Pile** is a poor predictor of downstream accuracy (Figure 30) though we note some models are trained on **The Pile** whereas others are not (Table 13). More broadly, given the many downstream results, we encourage

[23] See `https://crfm.stanford.edu/helm/v0.1.0/?group=disinformation_reiteration`.
[24] See `https://crfm.stanford.edu/helm/v0.1.0/?group=disinformation_wedging`.
[25] See `https://crfm.stanford.edu/helm/v0.1.0/?group=harms#Toxicity`.
[26] See `https://crfm.stanford.edu/helm/v0.1.0/?group=ablation_prompts`.
[27] See `https://crfm.stanford.edu/helm/v0.1.0/?group=ablation_multiple_choice`.